

Competition

Today's Agenda:

- Quiz
- Predator-Prey Interactions

Interspecific Competition

What do we mean when we say that two species compete?

Interspecific Competition

Occurs when one species limits the growth rate of another through:

- 1) Reducing resource availability (exploitative/resource competition)

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- 2) Reducing access to resources (interference/contest competition)
- 3) Increasing shared predators (apparent competition)

Interspecific Competition

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- 3) Increasing shared predators (apparent competition)

Can anyone think of examples of any of these?

Modelling Competition

Lotka-Volterra models again

Modelling Competition

$$\frac{dN_1}{dt} = r_1 N_1 \left(\frac{k_1 - N_1 - \alpha_{12} N_2}{k_1} \right)$$

dN_1/dT = rate of change in N_1

k_1 = carrying capacity of species 1

N_1 = # species 1

N_2 = # species 2

α_{12} = relative impact of sp. 2 on sp. 1

r_1 = growth rate of species 1

Modelling Competition

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$$\frac{dN_2}{dt} = r_2 N_2 \left(\frac{k_2 - N_2 - \alpha_{21} N_1}{k_2} \right)$$

dN_2/dT = rate of change in N_2
 k_2 = carrying capacity of species 2
 α_{21} = relative impact of sp. 1 on sp. 2
 r_2 = growth rate of species 2

Modelling Competition

α_{ij} = competition coefficient

$$\frac{dN_1}{dt} = r_1 N_1 \left(\frac{k_1 - N_1 - \alpha_{12} N_2}{k_1} \right)$$

$$\frac{dN_2}{dt} = r_2 N_2 \left(\frac{k_2 - N_2 - \alpha_{21} N_1}{k_2} \right)$$

dN_1/dT = rate of change in N_1
 k_1 = carrying capacity of species 1
 N_1 = # species 1
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 α_{12} = relative impact of sp. 2 on sp. 1
 r_1 = growth rate of species 1

dN_2/dT = rate of change in N_2
 k_2 = carrying capacity of species 2
 α_{21} = relative impact of sp. 1 on sp. 2
 r_2 = growth rate of species 2

Zero Net Growth Isoclines (ZNGIs)

- We can ask under what conditions populations are stable
- These are isoclines of zero net growth
- Set net growth to zero and solve for N_1

Zero Net Growth Isoclines (ZNGIs)

$$\frac{dN_1}{dt} = r_1 N_1 \left(\frac{k_1 - N_1 - \alpha_{12} N_2}{k_1} \right)$$

$$\frac{dN_2}{dt} = r_2 N_2 \left(\frac{k_2 - N_2 - \alpha_{21} N_1}{k_2} \right)$$

Zero Net Growth Isoclines (ZNGIs)

$$0 = r_1 N_1 \left(\frac{k_1 - N_1 - \alpha_{12} N_2}{k_1} \right)$$

$$0 = r_2 N_2 \left(\frac{k_2 - N_2 - \alpha_{21} N_1}{k_2} \right)$$

Zero Net Growth Isoclines (ZNGIs)

$$0 = \left(\frac{k_1 - N_1 - \alpha_{12}N_2}{k_1} \right)$$

$$0 = \left(\frac{k_2 - N_2 - \alpha_{21}N_1}{k_2} \right)$$

Zero Net Growth Isoclines (ZNGIs)

$$0 = k_1 - N_1 - \alpha_{12}N_2$$

$$0 = k_2 - N_2 - \alpha_{21}N_1$$

Zero Net Growth Isoclines (ZNGIs)

$$N_1 = k_1 - \alpha_{12} N_2$$

$$\alpha_{21} N_1 = k_2 - N_2$$

Zero Net Growth Isoclines (ZNGIs)

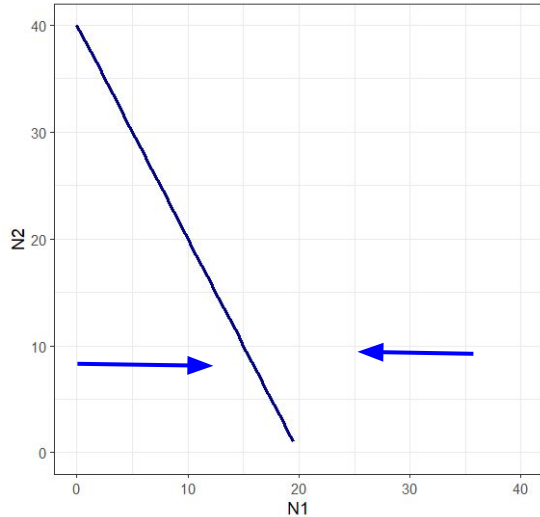
$$N_1 = k_1 - \alpha_{12}N_2$$

$$N_1 = \frac{k_2 - N_2}{\alpha_{21}}$$

Zero Net Growth Isoclines (ZNGIs)

Species 1

$$N_1 = k_1 - \alpha_{12} N_2$$

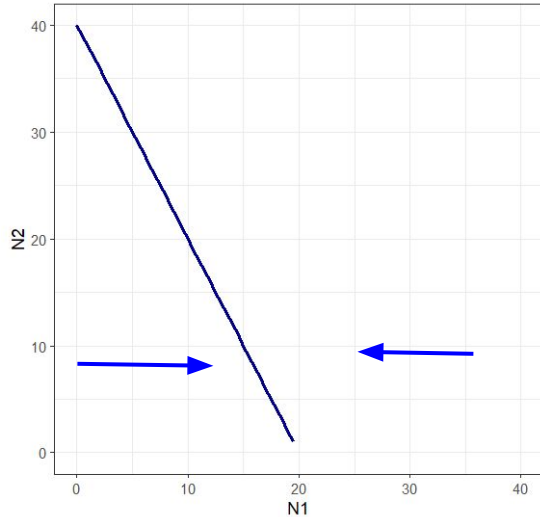


$$N_1 = \frac{k_2 - N_2}{\alpha_{21}}$$

Zero Net Growth Isoclines (ZNGIs)

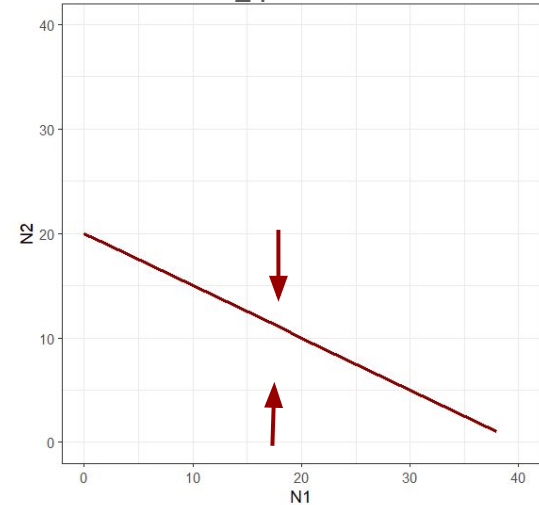
Species 1

$$N_1 = k_1 - \alpha_{12} N_2$$



Species 2

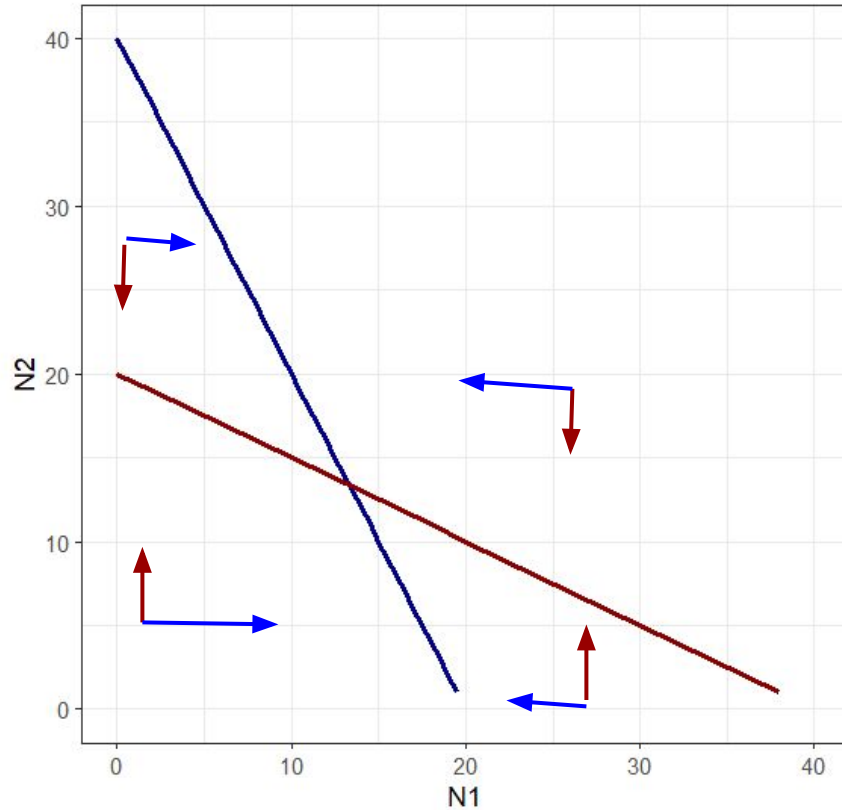
$$N_1 = \frac{k_2 - N_2}{\alpha_{21}}$$



Zero Net Growth Isoclines (ZNGIs)

Species 1

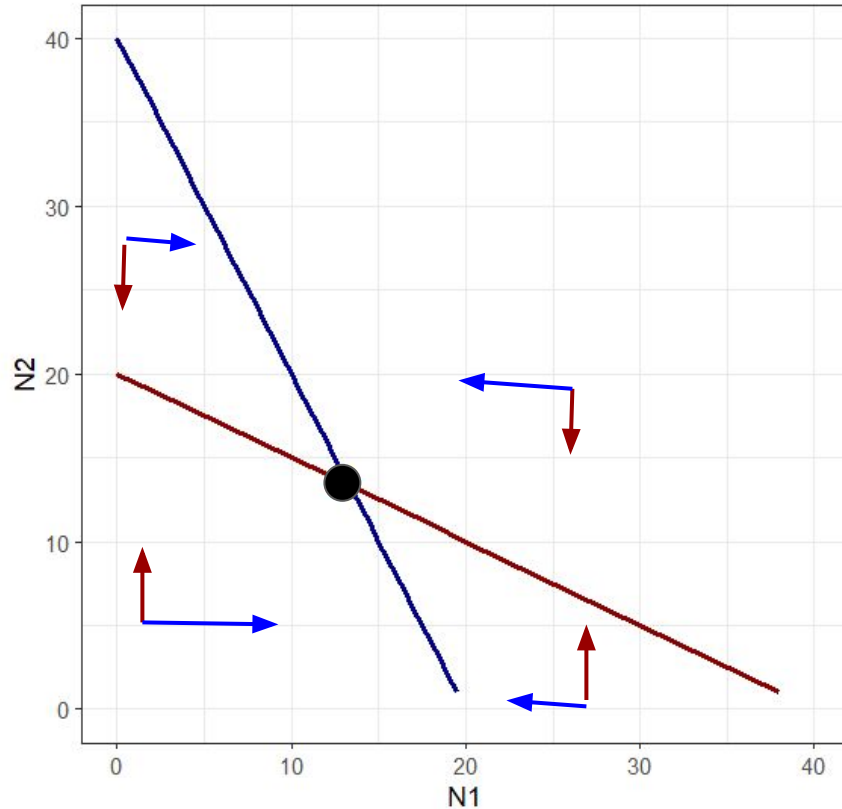
Species 2



Zero Net Growth Isoclines (ZNGIs)

Species 1

Species 2



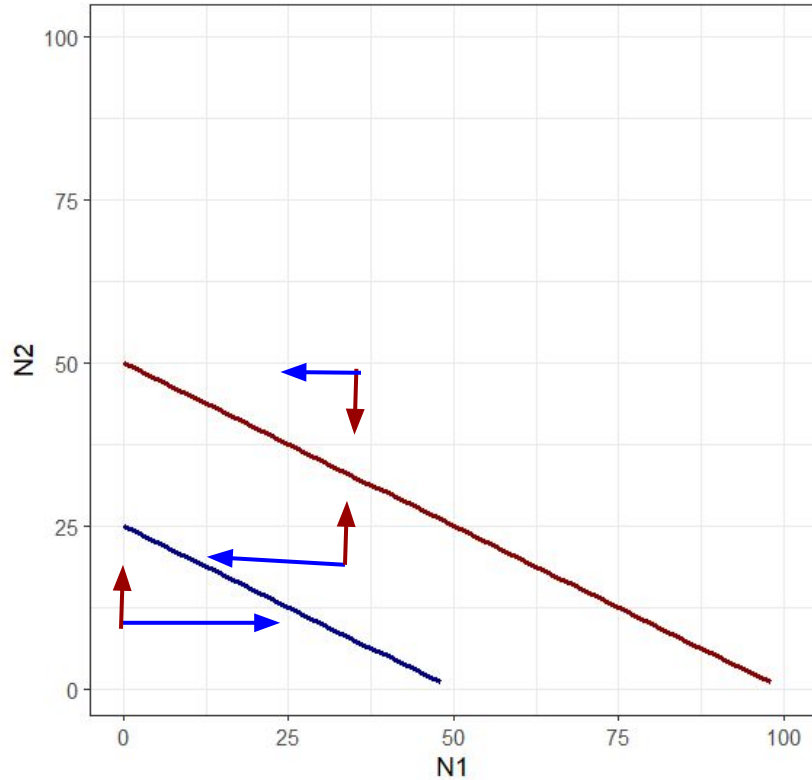
Stable coexistence

- Both species can invade from low density

Zero Net Growth Isoclines (ZNGIs)

Species 1

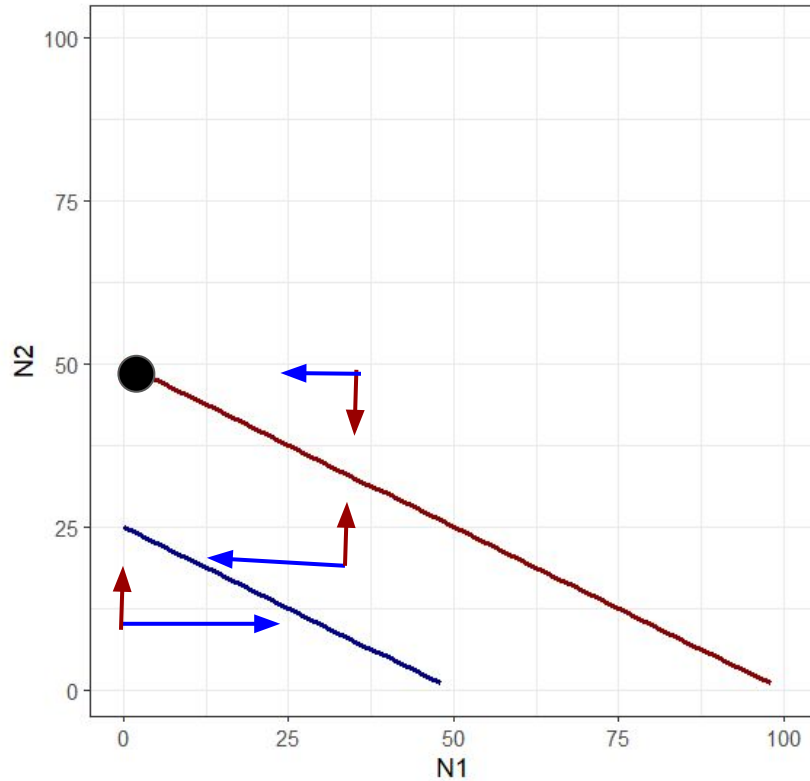
Species 2



Zero Net Growth Isoclines (ZNGIs)

Species 1

Species 2

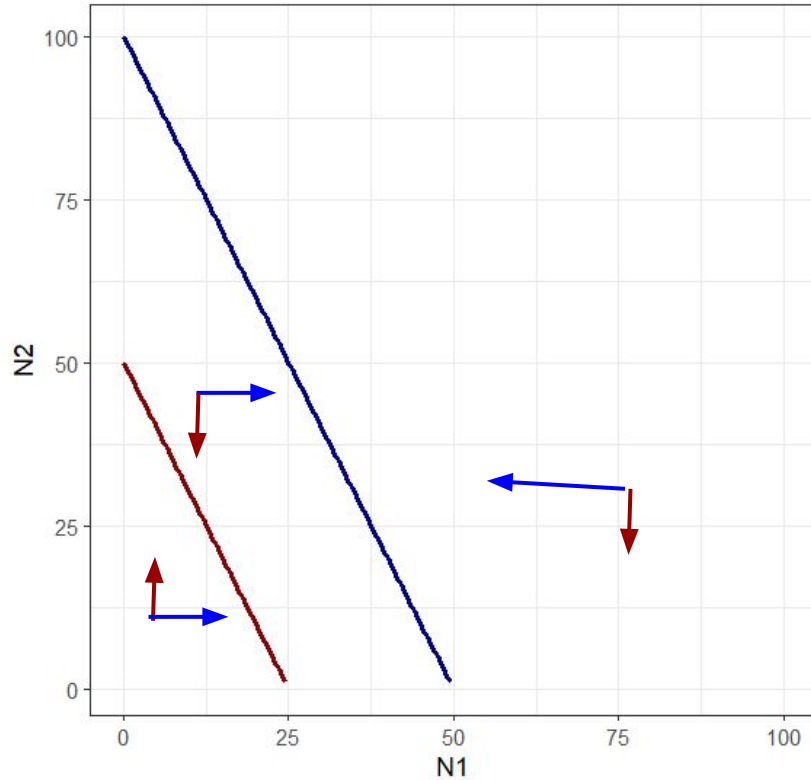


Species 2 excludes species 1

Zero Net Growth Isoclines (ZNGIs)

Species 1

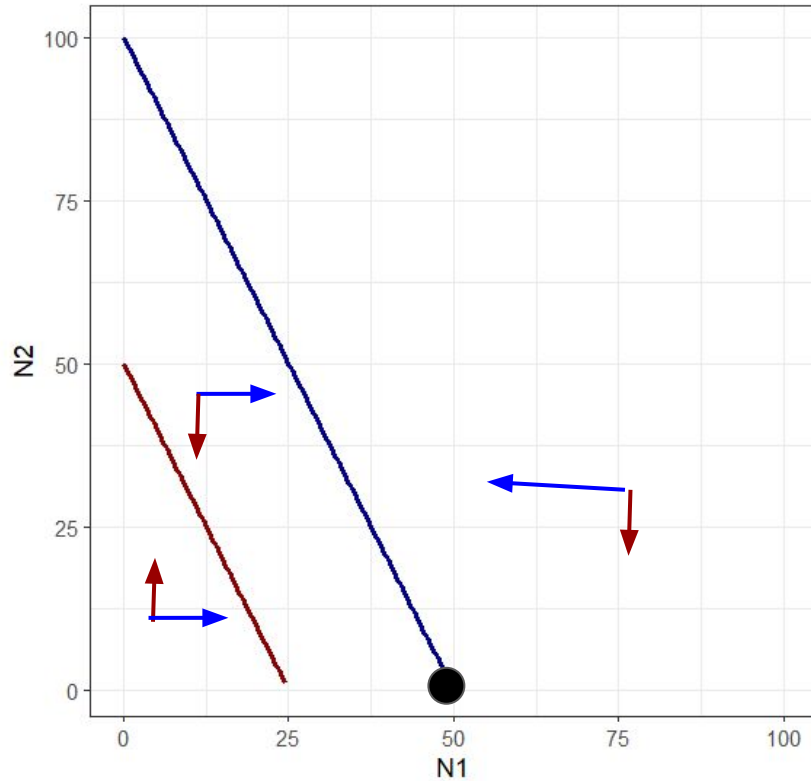
Species 2



Zero Net Growth Isoclines (ZNGIs)

Species 1

Species 2

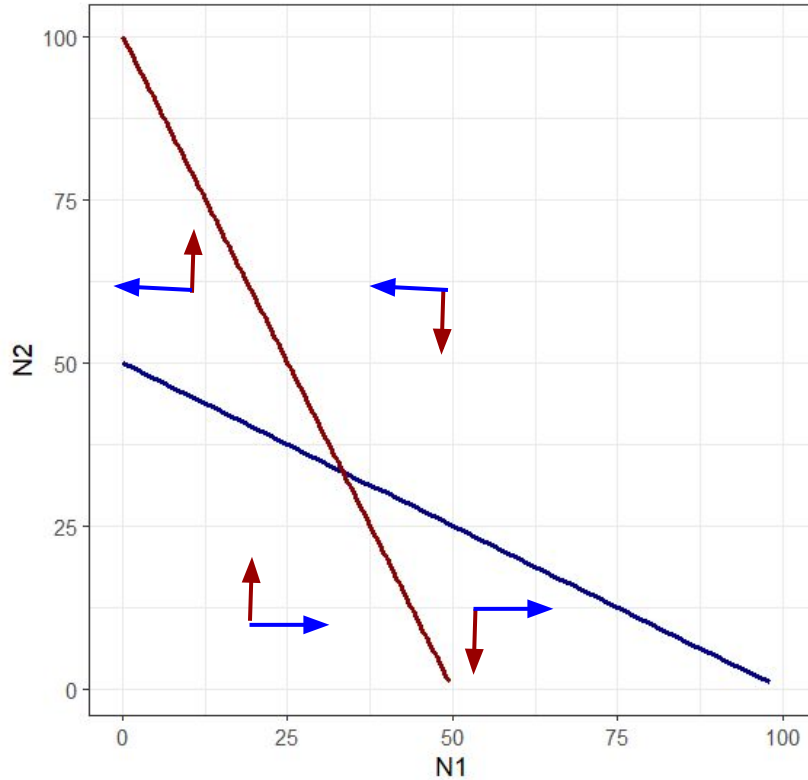


Species 1 excludes species 2

Zero Net Growth Isoclines (ZNGIs)

Species 1

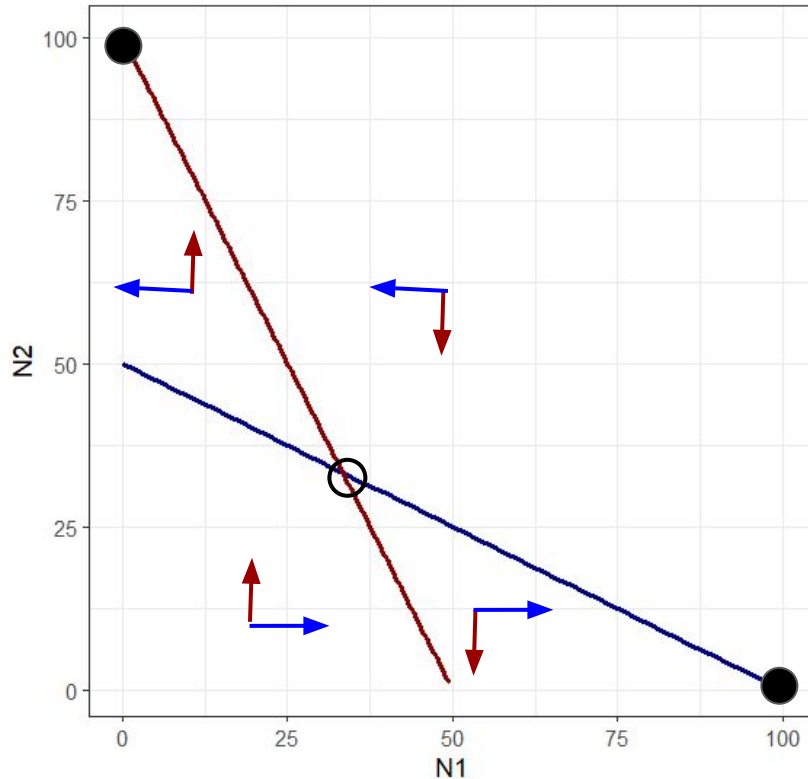
Species 2



Zero Net Growth Isoclines (ZNGIs)

Species 1

Species 2



ZNGIs can be confusing

Chesson 2000 provides a more intuitive formulation

$$\frac{1}{N_i} \cdot \frac{dN_i}{dt} = r_i (1 - \alpha_{ii}N_i - \alpha_{ij}N_j)$$

MECHANISMS OF MAINTENANCE OF SPECIES DIVERSITY

Peter Chesson

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Chesson 2000 provides a more intuitive formulation

$$\frac{1}{N_i} \cdot \frac{dN_i}{dt} = r_i (1 - \alpha_{ii}N_i - \alpha_{ij}N_j)$$

α 's here are absolute impact.

α_{ii} = intraspecific competition

α_{ij} = interspecific competition

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Coexistence criteria

Both species need to be able to invade when rare, when other species is present

$$\frac{1}{N_1} \cdot \frac{dN_1}{dt} = r_1 (1 - \alpha_{11}N_1 - \alpha_{12}N_2)$$

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Coexistence criteria

Both species need to be able to invade when rare, when other species is present

$$0 < r_1 (1 - \alpha_{11}N_1 - \alpha_{12}N_2)$$

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Both species need to be able to invade when rare, when other species is present

$$0 < 1 - \alpha_{12}N_2$$

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$$0 < 1 - \alpha_{12}N_2 \quad \frac{1}{N_2} \cdot \frac{dN_2}{dt} = r_2 (1 - \alpha_{22}N_2 - \alpha_{21}N_1)$$

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Both species need to be able to invade when rare, when other species is present

$$0 < 1 - \alpha_{12}N_2$$

$$0 = r_2 (1 - \alpha_{22}N_2 - \alpha_{21}0)$$

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$$0 = 1 - \alpha_{22}N_2$$

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Coexistence criteria

Both species need to be able to invade when rare, when other species is present

$$0 < 1 - \alpha_{12}N_2$$

$$\alpha_{22}N_2 < 1$$

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Both species need to be able to invade when rare, when other species is present

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$$N_2 = \frac{1}{\alpha_{22}}$$

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Coexistence criteria

In order to coexist:

$$\alpha_{ii} > \alpha_{ji}$$

$$\alpha_{jj} > \alpha_{ij}$$

α_{ii} = intraspecific competition

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In order to coexist:

$$\alpha_{ii} > \alpha_{ji}$$

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In other words: **to coexist, species need to limit themselves more than they limit their competitors**

α_{ii} = intraspecific competition

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**IMPORTANT:
WILL BE ON A TEST**



α_{ii} = intraspecific competition

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Resource competition

So far we've been talking about competition in terms of reduction in growth rate

- But competition usually isn't actually species interacting with each other
- It usually involves shared resources
- Resource models are more mechanistic

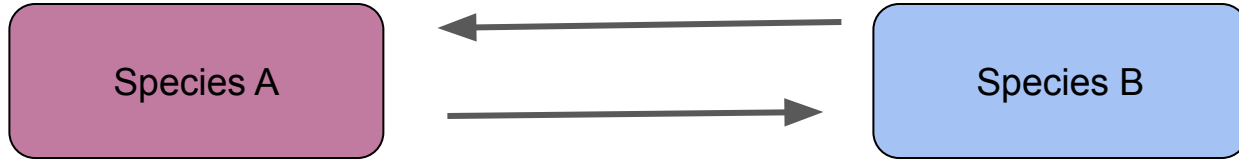
Resource competition



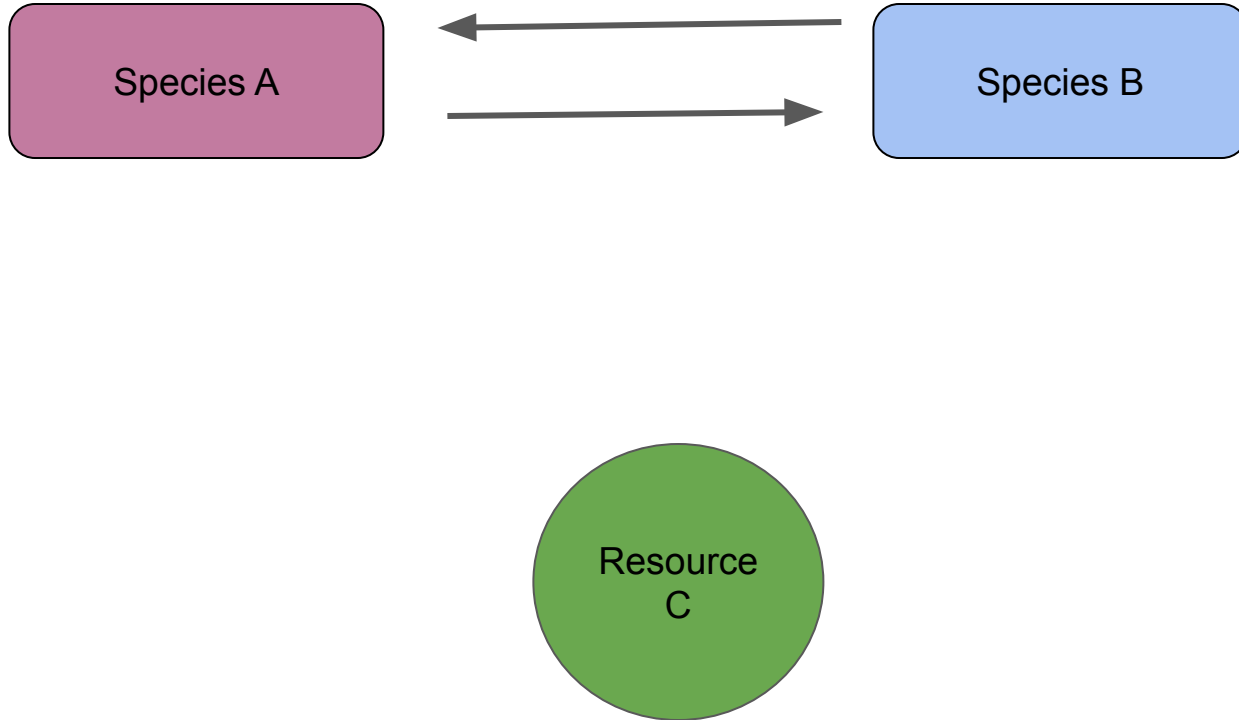
Species A

Species B

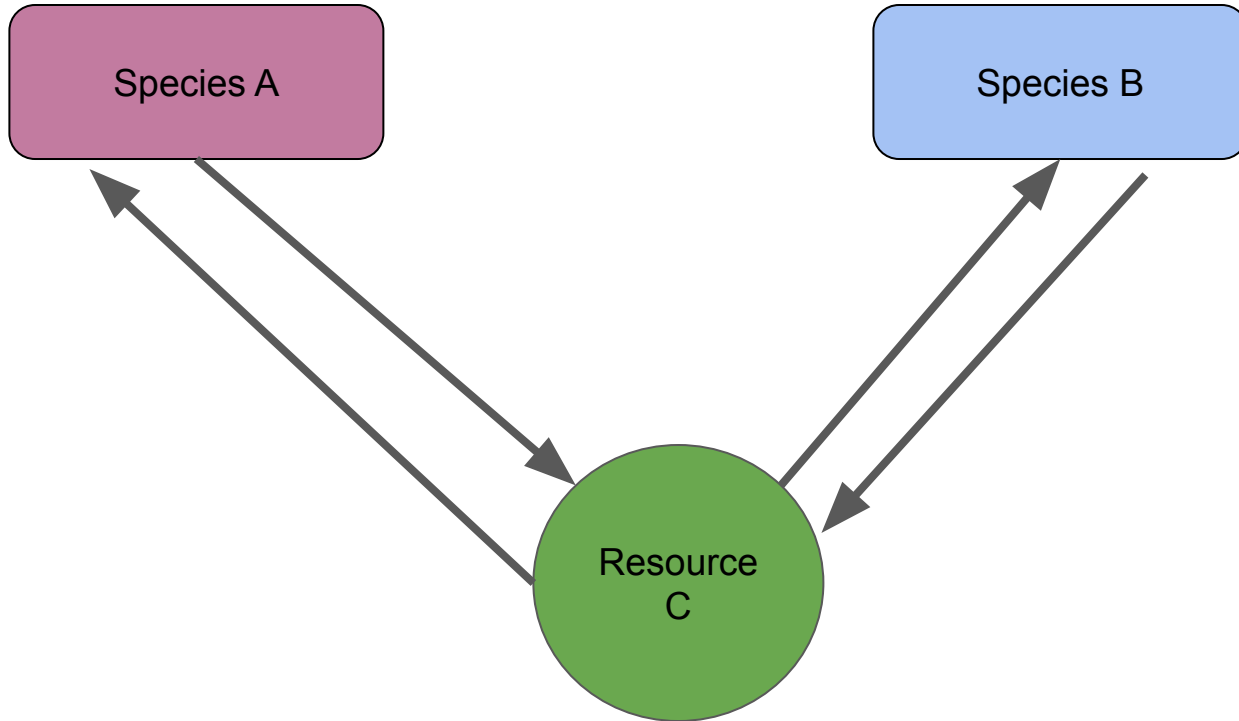
Resource competition



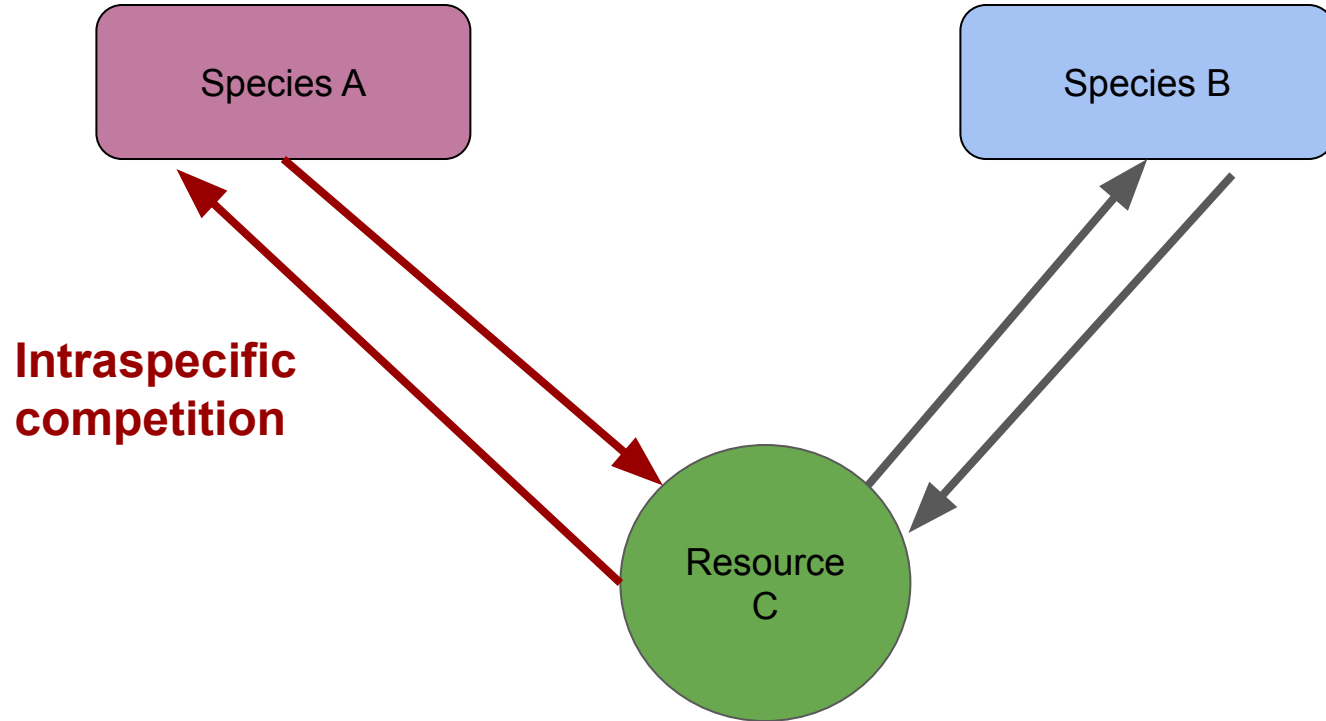
Resource competition



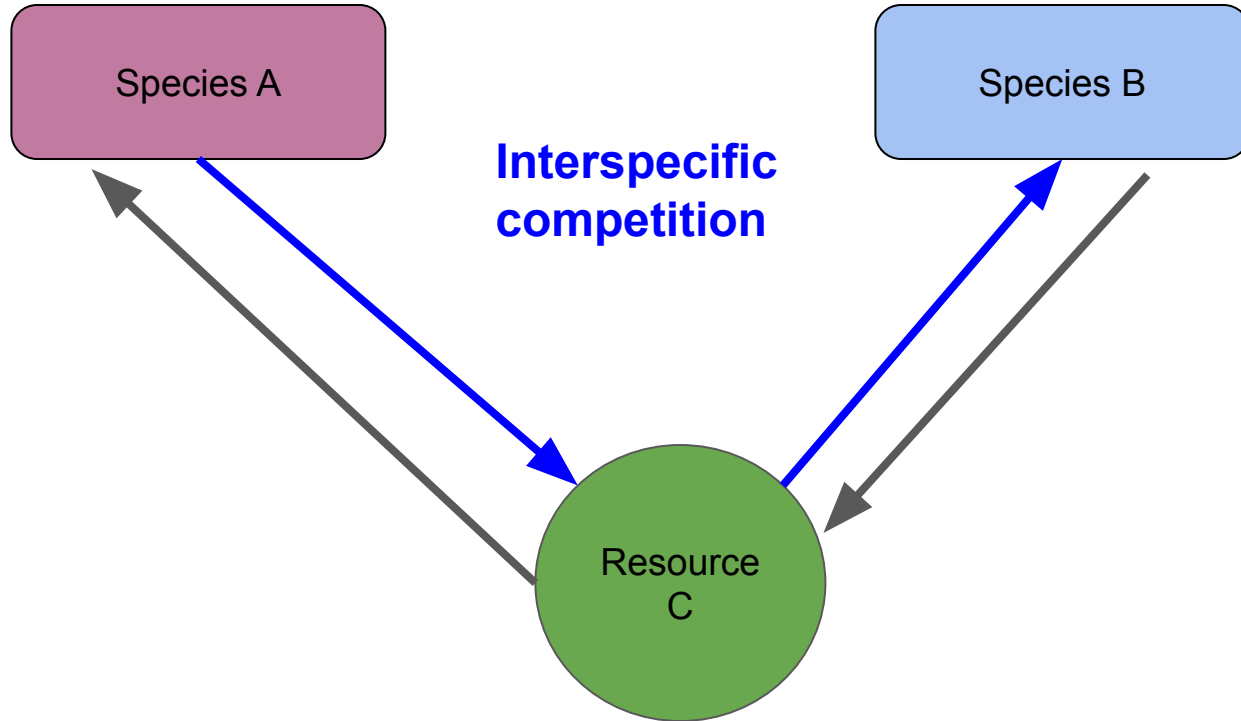
Resource competition



Resource competition



Resource competition



Resource competition models

What is a resource?

Resource competition models

What is a resource?

- Increases population growth
- Is consumed by use

Resource competition models

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Resources:

Food

Nutrients

Water (but trivial in aquatic)

Light (for producers)

Oxygen (but trivial in terrestrial)

Resource competition models

What is a resource?

- Increases population growth
- Is consumed by use

Resources:

Food

Nutrients

Water (but trivial in aquatic)

Light (for producers)

Oxygen (but trivial in terrestrial)

NOT Resources:

Temperature

Elevation

Pressure

R* Theory

$$\frac{1}{N_i} \cdot \frac{dN_i}{dt} = a_i R - d_i$$

R = resource availability

a_i = resource consumption rate of species i

d_i = death rate of species i

R* Theory

What is the minimum value of R needed?

$$\frac{1}{N_i} \cdot \frac{dN_i}{dt} = a_i R - d_i$$

R = resource availability

a_i = resource consumption rate of species i

d_i = death rate of species i

R* Theory

What is the minimum value of R needed?

$$0 = a_i R - d_i$$

R = resource availability

a_i = resource consumption rate of species i

d_i = death rate of species i

R* Theory

What is the minimum value of R needed?

$$\frac{d_i}{a_i} = R$$

R = resource availability

a_i = resource consumption rate of species i

d_i = death rate of species i

R* Theory

What is the minimum value of R needed?

$$R^* = \frac{d_i}{a_i}$$

R = resource availability

a_i = resource consumption rate of species i

d_i = death rate of species i

R* Theory

$$R^* = \frac{d_i}{a_i}$$

- Minimum resources required
- Will determine equilibrium abundance
 - Population will increase until it draws resources down to this level
- Determined by death rate, consumption rate

R = resource availability

a_i = resource consumption rate of species i

d_i = death rate of species i

R^* determines who wins in competition

$$R^* = \frac{d_i}{a_i}$$

The species with the lowest R^* wins

- Will draw resource down to a level that excludes all other species

R = resource availability

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- Will draw resource down to a level that excludes all other species
- Note this is only true for species limited by a single resource under constant conditions
- But is still useful generally

R = resource availability

a_i = resource consumption rate of species i

d_i = death rate of species i

Competition for multiple resources

- Gets a lot trickier
- Resources can be essential vs substitutable

Competition for multiple resources

- In order to stably coexist
 - Species need to differ in resource usage
 - Species need to have a larger impact on the resources that are most limiting for them

Competition for multiple resources

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These conditions lead to **species limiting themselves more than they limit their competitors** ($\alpha_{ii} > \alpha_{ji}$)

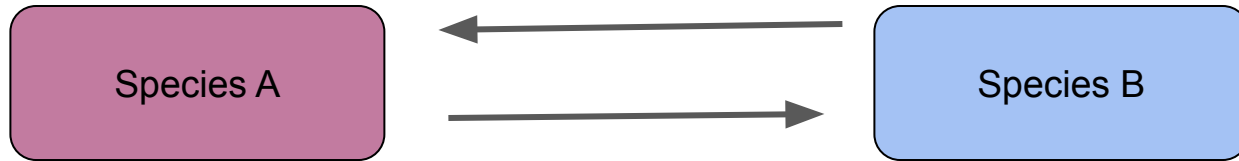
Competition for multiple resources

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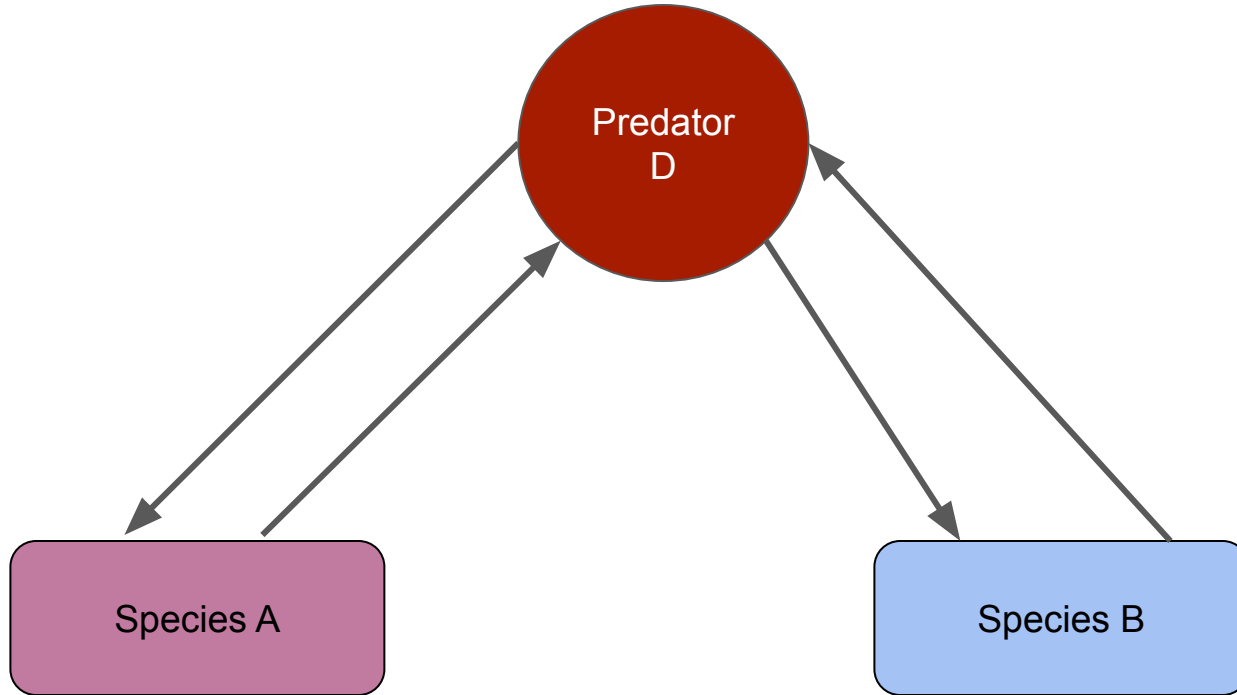
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When coexistence is purely driven by resource differences, the number of species that can be supported is equal to the number of different resources

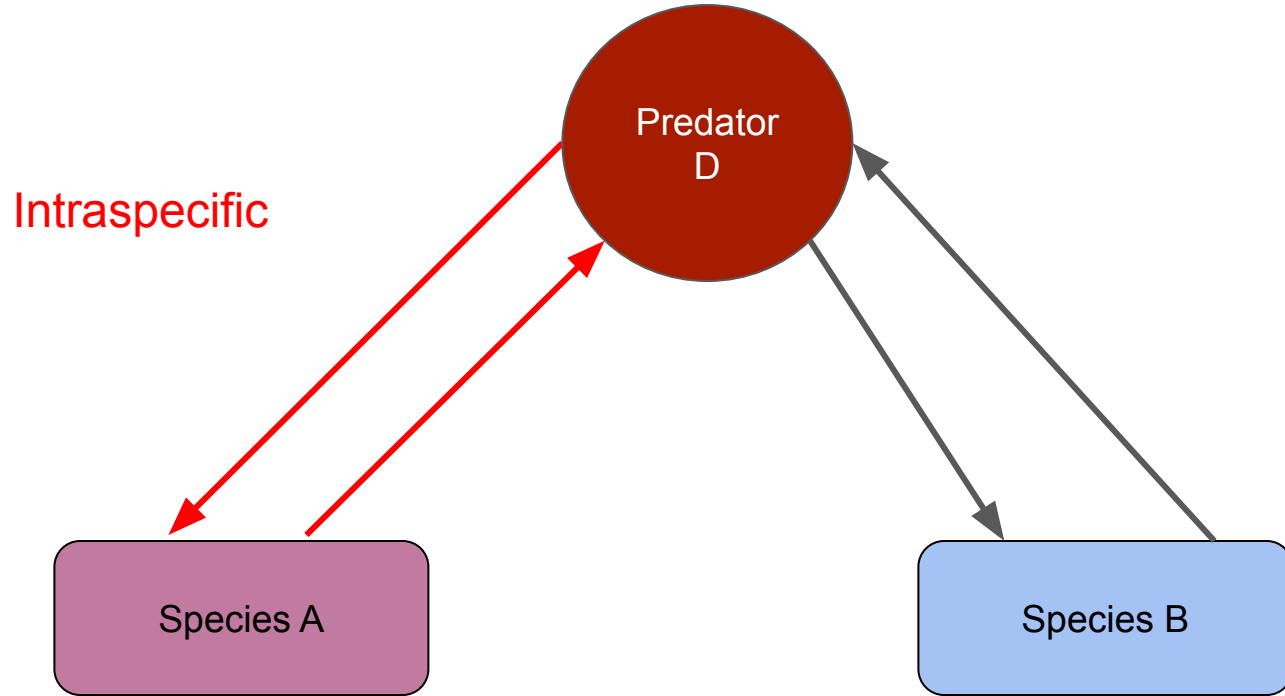
Apparent Competition



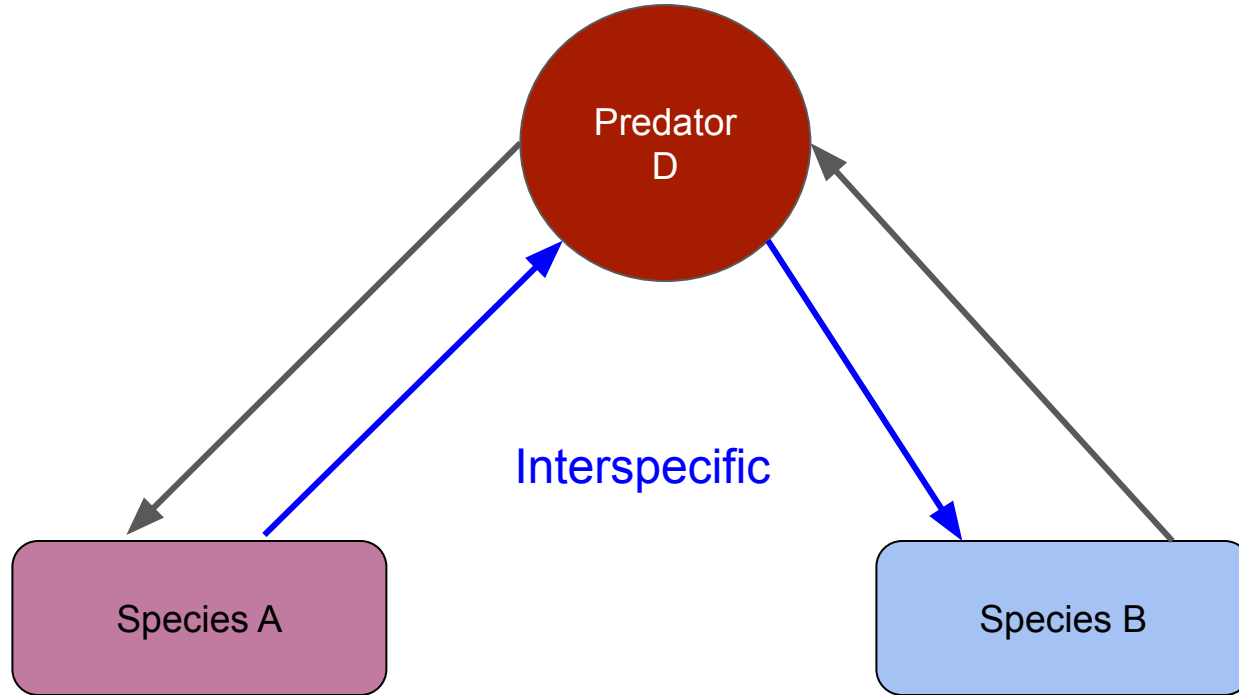
Apparent Competition



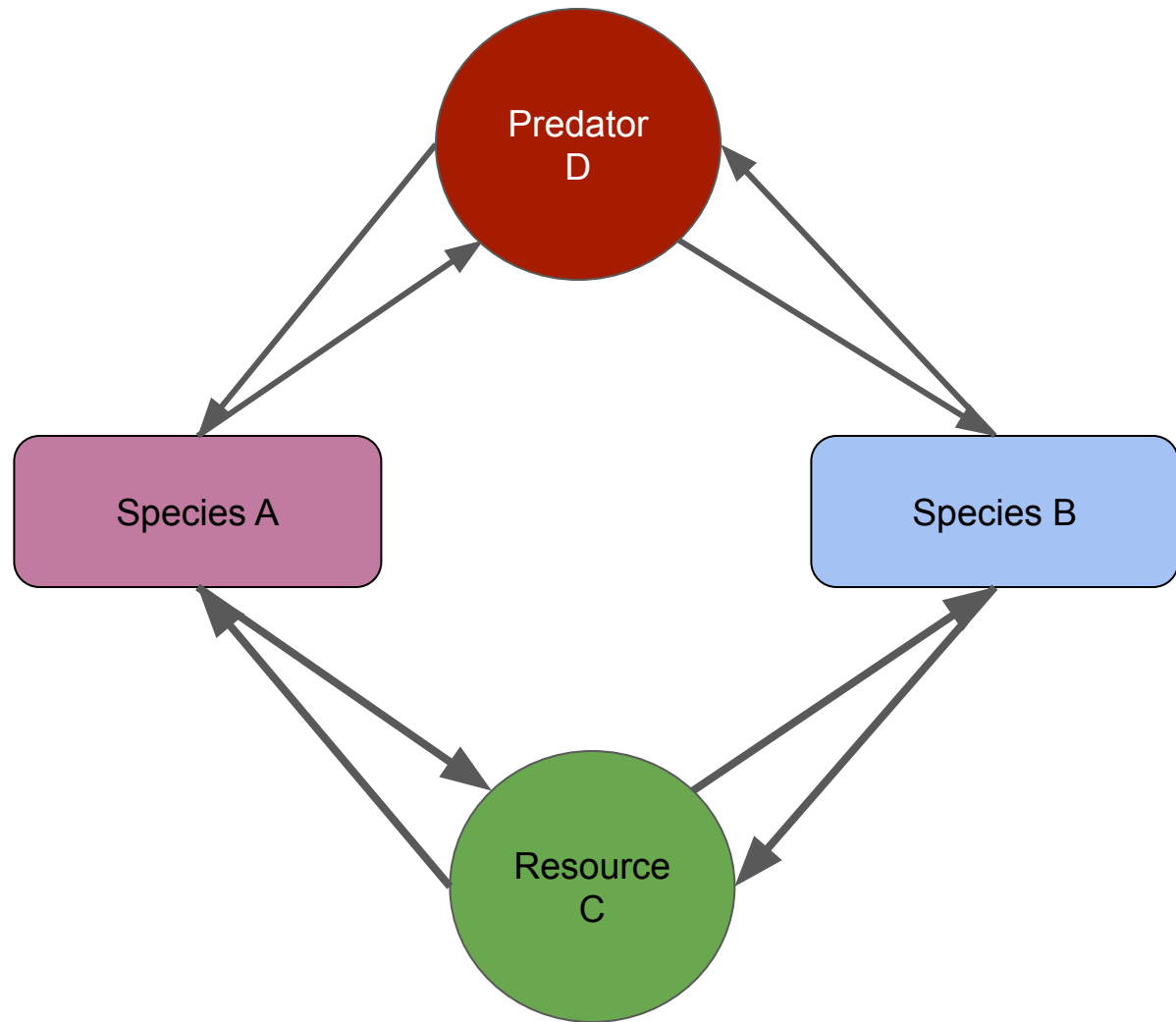
Apparent Competition



Apparent Competition



Competition

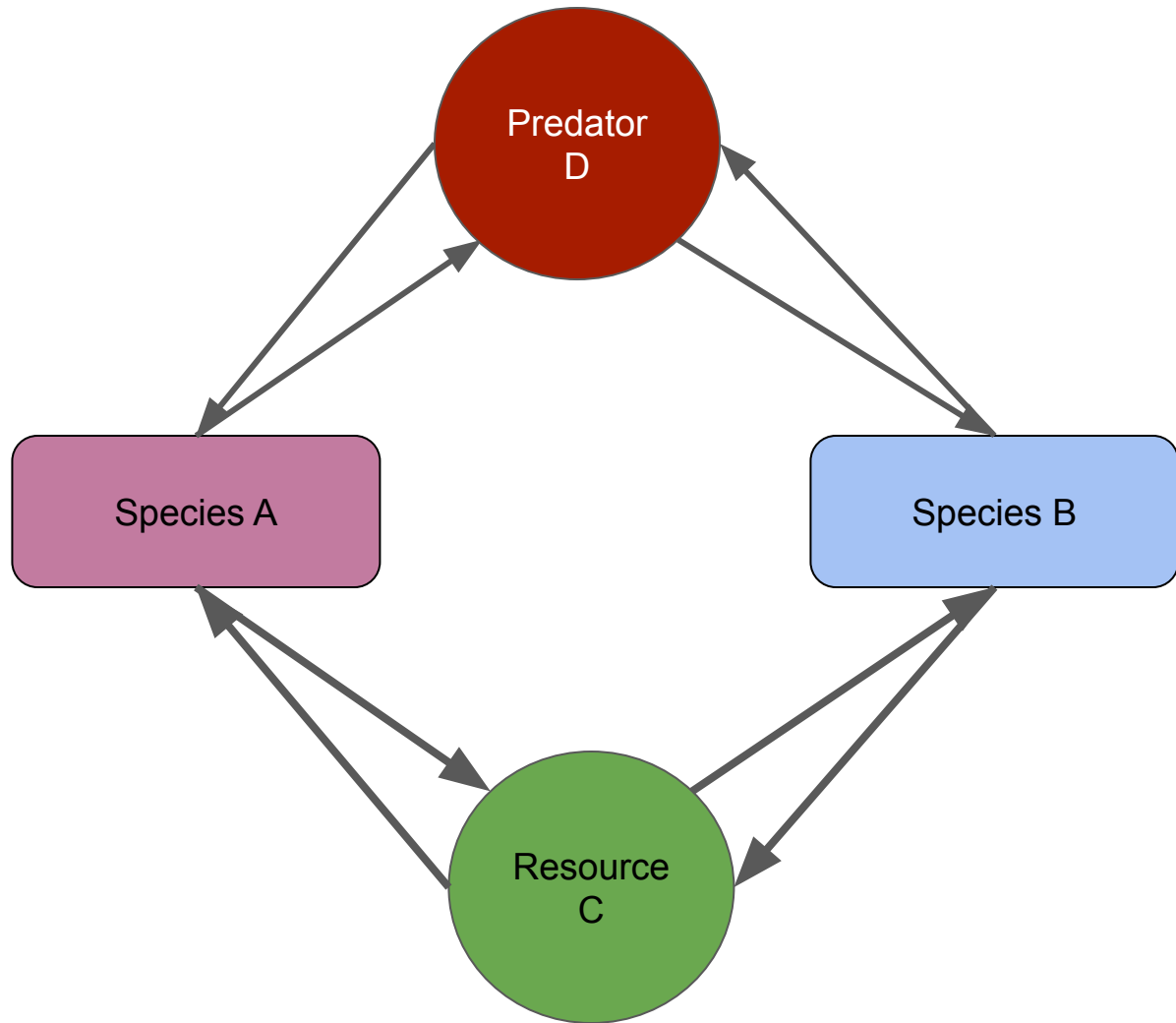


Competition

Competition and Predation
are symmetrical!

Promote coexistence in the
same ways!

2 species can coexist with
2 predators, 2 resources,
1 predator and 1 resource



P^* determines who wins in apparent competition

$$P_i^* = \frac{r_i}{a_i}$$

P^* is symmetrical with R^*

The species with the *highest* P^* wins

- Will increase predation to a level that excludes all other species
- Note this is only true for species limited by a single resource under constant conditions
- But is still useful generally
- This version is for Type I functional response

P^* = equilibrial predator density

r_i = intrinsic growth rate of species i

a_i = predator **attack rate** on species i

Main Takeaways

- To coexist, species need to limit their own growth more than their competitors
- Species that can exist on the lowest amount of a resource will win
- Species that can tolerate the highest level of a given predator will win
- Predation is symmetrical with competition
- Resources and predator diversity determines # of coexisting competitors

Next class: Coexistence